Chapter 10 Warm-Up: Dictionaries and a Phone Book

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1 Dictionary Basics

1. A Python dictionary stores information as **key–value pairs**. Consider

```
phone_book = {
 "562-555-0142": "Maria Garcia",
 "562-555-0188": "Daniel Kim",
 "562-555-0125": "Elena Garcia",
 "562-555-0171": "Maria Lopez",
 "562-555-0194": "Aisha Patel"
}
```

   Each phone number is a **key**. The corresponding name is its **value**. What is printed by

```
print(phone_book["562-555-0188"])
```

   Why is it useful for each dictionary key to be unique?

2 Creating and Adding Entries

1. An empty dictionary can be created using

```
phone_book = {}
```

   A new key–value pair can then be added:

```
phone_book["562-555-0142"] = "Maria Garcia"
```

   Add another entry:

```
phone_book["562-555-0188"] = "Daniel Kim"
```

   Print the entire dictionary:

```
print(phone_book)
```

   What happens if you assign a new value to a key that already exists?

3 Dictionary Operations

1. The method `phone_book.keys()` returns the dictionary keys. The method `phone_book.values()` returns the values. The method `phone_book.items()` provides the key and value together. For our phone book `(phone, name)` represent one key–value pair.

4 Searching by Name

1. We can combine dictionaries with the string-searching techniques from Chapter 8. Begin with

```
search = input("Enter first/last name, or both: ")
```

   Then examine each entry:

```
for phone, name in phone_book.items():
    if search.lower() in name.lower():
        print(name, phone)
```

   Predict the output when the user enters

```
Garcia
```

   How many phone numbers should be printed?

5 Put the Search in a Function

1. Write a function that returns all matching phone numbers:

```
def find_numbers(phone_book,search):
    matches = []
    ...
    return matches
```

   Test the function:

```
print(find_numbers(phone_book,"Maria"))
```

   Which phone numbers should be returned?

6 Warm-Up Exercise

Using the original five-entry dictionary, predict the result of each search before running the program:

```
find_numbers(phone_book,"Garcia")
find_numbers(phone_book,"Maria")
find_numbers(phone_book,"Daniel Kim")
find_numbers(phone_book,"Johnson")
```

Answer the following:

- Which search returns two phone numbers?
- Which search uses both a first and last name?
- What does the function return when there are no matches?
- Why is a list used for `matches` instead of returning the first phone number immediately?